

USING PAST EXPERIENCE IN MODERN ANALYSIS OF ARCH DAMS

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ABSTRACT

There are about 2,300 concrete arch dams built in the World that are of different shapes ranging from very thin dams to thick curved gravity structures. These structures were, built with various materials and construction technologies, and are designed for different loads using variety of analysis methods. Among these methods, the Trial-Load Method was the prevailing approach in the design of arch dams for over half of the 20th century. The method was verified and validated in 1920's and was commonly accepted in the engineering practice.

In the 1960's, Trial-Load Method was computerized by the US Bureau of Reclamation and the Arch Dam Stress Analysis System, known as ADSAS program, significantly shortened the analysis process. As the Finite Element Method was more commonly used by engineers in 1980s, the Trial-Load Method and the ADSAS program started losing its significance.

The authors of the paper will present an updated version of ADSAS program. The new 2024 ADSAS program was compiled using the original Reclamation's Fortran code and a new graphical interface was added to facilitate the input and the visualization of the results including displacements, principal stresses on the dam faces, and loads on cantilevers and arches. The authors' intention is to make the program available for educational purposes, to teach younger engineers and students the basic principles of arch dam analysis and design, and to preserve this important piece of engineering history.

In the paper, the authors will present a verification process of the ADSAS results using a software-to-software method and inclusion of the 60-foot-high Stevenson Creek Experimental Dam, as an example of an arch dam model, constructed and used exclusively for validation of the Trial-Load Method.

INTRODUCTION

The Trial-Load Method was the prevailing approach in the design of arch dams for over half of the 20th century. The method was verified and validated in 1920's and was commonly accepted in the engineering practice in design of arch and gravity dams. In the 1960's, the Trial-Load Method was computerized by the US Bureau of Reclamation (Reclamation) and the Arch Dam Stress Analysis System, known as ADSAS program, significantly shortened the analysis process. As the Finite Element Method (FEM) was more commonly used by engineers in 1980s, the Trial-Load Method and the ADSAS program started losing its significance.

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An intention of this paper is to present the ADSAS program with a new more user-friendly graphical interface that can be used for educational purpose and to keep this historical great piece of engineering achievement alive. Although the Trial-Load Method is used by few practitioners in the current engineering practice, it is remaining a part of university programs educating future dam engineers. Its intuitive formulation allows the students and younger engineers easier understand a general concept of arch dam behavior.

In the paper, general principles of arch dam behavior and a brief historical overview of the Trial Load Method and the ADSAS program developments are presented. In conjunction, a new version of ADSAS with a user-friendly interface and a software-to-software verification procedure, using a Stevenson Creek Experimental Dam model, for build confidence in the analysis results is also presented.

ARCH DAM BEHAVIOR

The traditional engineering practice has been based on a concept of an “Arch and Cantilever” approach for explanation of load distributions in arch dams. However, in earlier construction of arch dams at the turn of the 19th century, dam shapes and their thickness were determined using a thin-cylinder formula. For more context, a historical perspective of the development of analysis methods used for layout of arch dams is presented by Tarbox, Nuss and Salamon (Tarbox et.al, 2020).

In the Arch and Cantilever concept, the loads are transferred to the foundation by the horizontal arch and vertical cantilever elements, producing equal arch and cantilever deflections in radial directions. The proportion of loads distributed between arches and cantilevers depends primary on the curvature and thickness of a dam, with dominating arch action in thin arch dams and significant cantilever element contribution in thick arch dams. In reality however, embedment of the dams in abutments creates bending moments and shear forces in arch elements resulting in:

- An “arch dropping” effect at abutments.
- A reorientation effect of the resultant forces in arche sections toward downstream.
- Bending moments in the arch dam section as an effect of the dam deformation and restrains at abutments and the foundation.

The readers can find more information describing various concepts of arch dam behavior in several publications (ASCE 1957, Cain 1922; Copen 1960, ICOLD 2024; Jargensen 1914, Kollgaard 1988, Noetzili 1921, Rydzewski 1964; Tarbox et.al 2020, USBR 1977a, USBR 1977b).

AN HISTORICAL OVERVIEW OF THE TRIAL LOAD METHOD

The Trial-Load Method is an advancement of the Arch and Cantilever concept that has been intensively developed by Reclamation for over a half-century, starting in 1923 (USBR 1938; USBR 1977a; Tarbox et.al, 2020). The method is an iterative process where loads are adjusted between all arch and cantilever intersections until the specified criterion is met, or “Trial Load” Method. Although the concept of the Trial Load Method was used in the design of several dams since the beginning of the 20th century, the actual name “Trial-Load Method” was first publicly used by Howell and Jaquith in their 1929 paper (Howell, 1929).

In the Trial-Load Method, loads are distributed between the number of vertical cantilever and horizontal arch elements. The division of load does not need to be uniform, but the final division is one that results in equal deflections in radial, tangential, and rotational directions for all points of intersection between the cantilever and arch elements (instead of an agreement between the arches and just the single crown cantilever in the Arch and Cantilever method) and each arch and cantilever elements are assumed to move independently of all others. At the conclusion of the analysis each arch and cantilever element must remain in equilibrium and geometrical continuity at all intersection points and important to note that the sum of the arch and cantilever loads must equal the external load (weight, reservoir, tailwater, temperature, silt, and ice) at each intersection point. These external loads are divided between arches and cantilevers to get satisfactory radial deflection agreement. To complete the deformation agreement without changing the external loads, internal pairs of equal and opposite tangential and twist loads are applied to arches and cantilevers. Computations of deflections of arch elements are made by the elementary theory of flexure for curved beams with the effects of rib-shortening, transverse shear, and yielding abutments.

Overall, the Trial Load Method produces:

- cantilever stresses vertically along all cantilevers;
- arch stresses at multiple locations along all arches;
- principal stresses in the correct orientation not just horizontal along the arches; and
- displacements and rotations at all cantilever and arch intersections.

ADSAS: THE COMPUTERIZED TRIAL-LOAD METHOD

Trial-Load Method was computerized in the 1960s by Reclamation. The Arch Dam Stress Analysis System (ADSAS) computer program greatly advanced the state-of-the-practice for arch dam analysis, speed up the design process, and helped justify the use of engineering mainframe computers. ADSAS changed the way the Concrete Dam Group at Reclamation operated because more load combinations and geometrical shapes could be investigated in minutes rather than weeks.

The computerization of the Trial-Load Method greatly helped in the development of doubly curved arch dams because many analyses could be quickly performed. In the 1950s, before ADSAS, one complete trial load analysis took 6 to 12 months for one senior engineer to do the calculations and one junior engineer to carry out a simultaneous check. With the speed of ADSAS, the final arch shape for Morrow Point Dam, for example, was obtained after 60 iterations for gravity, reservoir, silt loads, and winter/summer temperature conditions. Despite the advances that came with ADSAS, it was not appropriate for dynamic analysis (Tarbox et.al, 2020).

ORIGINAL ADSAS VERSION

The original version of ADSAS program and the user's manual was developed by Reclamation for internal use; there was a machine-dependent computer code specifically for a Cyber 70-74/28 computer, and the program required more than 39,000 cards long with over 240 subroutines. This caused problems for users to transferring the program to their computers, as well as using the program. Output from ADSAS was originally in a paper form and was about 1 to 2 inches thick when completed. Designers would thumb through the large volume of the paper output, make some hand calculations, propose changes to the dam geometry, and then have the younger engineers run ADSAS, returning the revised paper output for review (Tarbox et.al, 2020).

Later in the 1990's US Army Corps of Engineers (USACE, 1997) provided a MS-DOS executable ADSAS version allowing a user to run the analyses using numerical input files such as the example in Figure 1. The input described the required data for the analysis: dam geometry, material properties, loads and several other parameters used in computations. An output text file then provided a summary of the analyses and the results. This executable file can still be run but is not very practical compared with modern computation tools with a more user-friendly interface.

```

- NAME=HAYO 111011111
- 1 0 ADSAS Test Problem 14 01
- 2 0 011011101
- 3 0 30100000000 3.0 .2 5.6 150. 2.5 .2 .0 .0
- 4 0 6700. 7165. 7165. 6700. 7165.
5 1 1 1 34.427 17.227 221.1513 174.6713 375.0 .0 3 1
5 1 2 1 7165.0 .0 772.0 6874.069 825.0
5 1 2 2 6950.0 .0 947.3946 6893.103 950.0
5 1 2 3 7165.0 .06 .0 .0 .0
5 1 3 1 7000.0 .0 846.4621 6598.577 700.0
5 1 3 2 7041.622 .7 .0 .0 .0
5 1 3 3 7100.0 .0 -421.5882 7548.229 -883.4174
5 1 3 4 7160.0 .5888 .0 .0 .0
5 1 3 5 7165.0 .0 .0 .0 .0
5 1 4 1 7103.008 .0 1289.338 6544.889 1100.0
5 1 4 2 7160.0 .5888 .0 .0 .0
5 1 4 3 7165.0 .0 .0 .0 .0
6 1 .0 12.0 .0 .0 .0 .0 .0
7 110000 6700.0 .0 .0 .0 .0 .0 .0
7 100000 6790.0 44.350 44.350 .0 .0 -1.5 .0 .0
7 100000 6865.0 48.125 48.125 .0 .0 -1.5 .0 .0
7 100000 6940.0 49.600 49.600 .0 .0 -1.5 .0 .0
7 100000 7015.0 50.250 50.250 .0 .0 -2.0 .0 .0
7 100000 7090.0 51.625 51.625 .0 .0 -4.5 .0 .0
7 100000 7165.0 54.875 54.875 .0 .0 -15.0 .0 .0
- 1 2 1 1 111 1
- 1 4 1 1 1 1 1 1 1
- 1 3 1 1 1 1 1 1 1
- 10005 1 1 1 1 1 1 1 1
- 1 10 1 1 1 1 1 1 1
*ENDFILE

```

Figure 1. An example of an input for original ADSAS version

UPDATED ADSAS VERSION FOR EDUCATIONAL PURPOSE

The authors recompiled the original Reclamation's code using Fortran 94 compiler and added a graphical interface for:

- Easy ADSAS installation and runs on personal computers with Windows OS;
- Easier data input for defining dam models;
- Direct and easy visualization of the analysis results (displacements, principals stresses, forces);
- User-friendly comparison of several results computed for different load conditions; and
- Conversion between Imperial units and IS units.

In this 2024 version of the updated ADSAS program, only single-center and two-center layouts are available (a three-center layout that required modelling of more complex geometry will be added in the future). In the new interface, the user defines the geometry of the central cantilever (upstream and downstream face) using straight line or circular segment. In a similar way, upstream and downstream line of center needs to be defined. In a second step, at several elevations, angles from the reference plane to the abutment end of the arch are used to define not only the geometry of the abutment but also the number and location of arches and cantilevers (Figure 2).

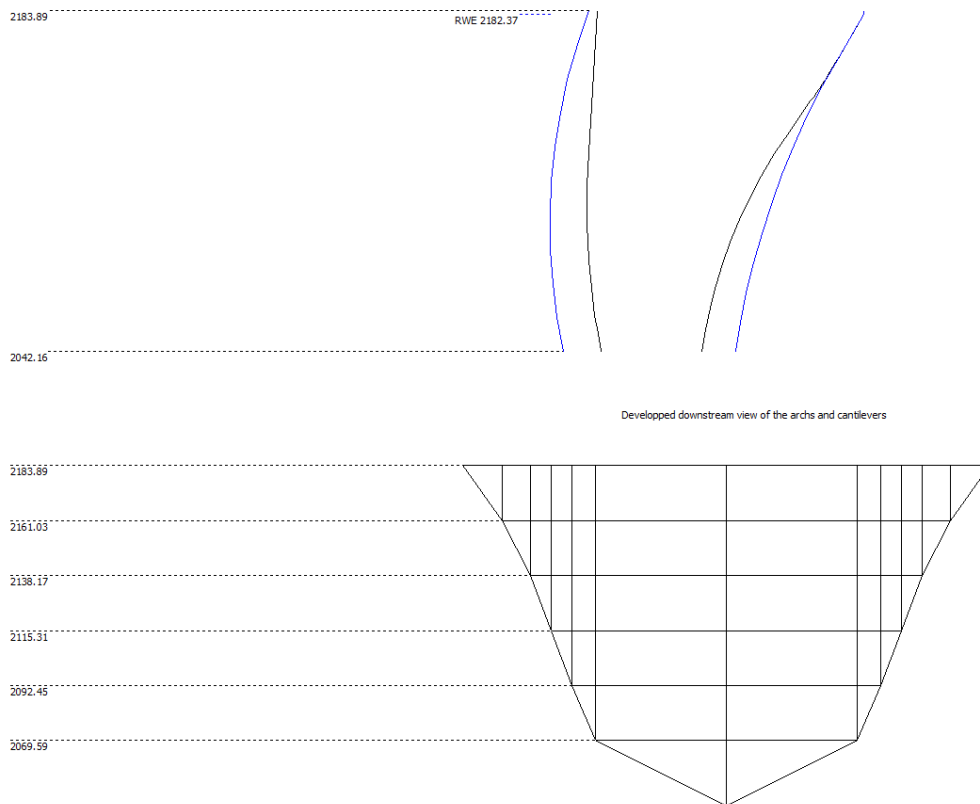


Figure 2. Geometry description: us/ds face of central cantilever and us/ds line of center (top), arches and cantilevers (down)

The input parameters include material properties (modulus of elasticity of concrete and rock, Poisson's ratio, and unit weight of concrete), and loads (reservoir and tailwater water elevations, temperature condition). As the analysis is run, several graphical results are directly available (Figure 3):

- Radial displacement at each node between arches and cantilever;
- Global and principal stresses on the upstream and downstream faces: blue (compression) and red (tension) allows to directly 'understand' the stress pattern;
- Orientation and value of the forces on the abutment; and
- Distribution of the hydrostatic load between arches and cantilevers.

These results obtained from ADSAS are considered representative of the typical data used in design of arch dams. With a quick change of any parameter describing the dam geometry, material or loads, new results can be computed within a few seconds. This allows the user to perform multiple analyses to understand the influence of each input parameter on the analysis results.

Figure 4 shows the principal stresses on the upstream and downstream faces of a Stevenson Creek Experimental Dam, resulting from an evaluation by Salamon & Robbe (Salamon et.al. 2023) for 3 load cases:

- Self-weight load only.
- Self-weight + hydrostatic loads.
- Self-weight + hydrostatic + 10°C temperature increase loads.

From the vertical only stress state under self-weight load only, the stress pattern is strongly modified under hydrostatic load:

- Horizontal compressive stresses are computed on the upstream and downstream face at the central cantilever, gradually bending toward the abutment.
- Tensile stresses normal to the abutment are computed near the upstream toe.
- Vertical tensile stresses are computed on the downstream face.

Thermal load also strongly influenced the stress pattern: concrete expansion leads to redirect forces toward the abutments.

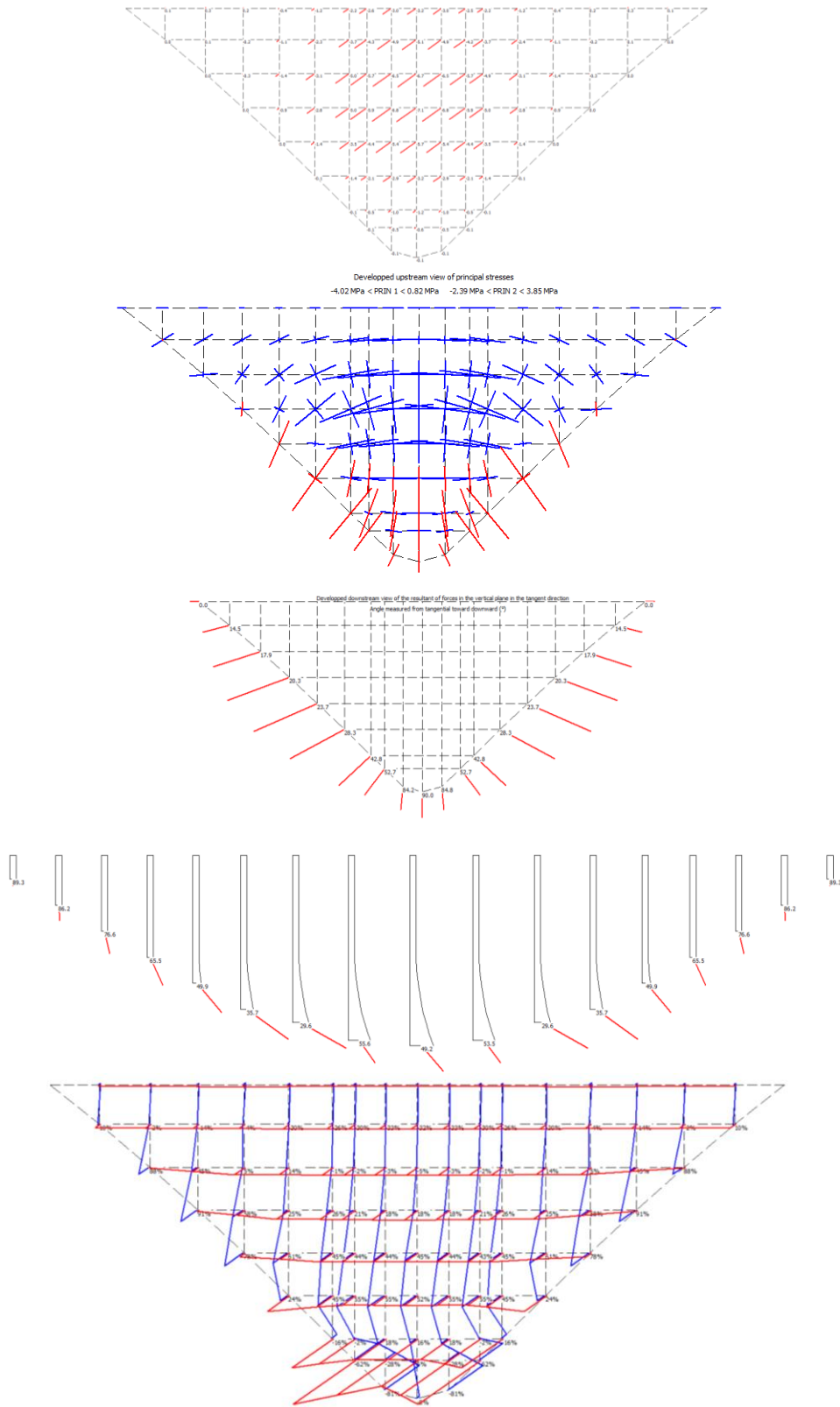


Figure 3. Example of ADSAS results - from top to bottom: radial displacements, principal stresses on upstream face, forces on abutment, arch-cantilever loads

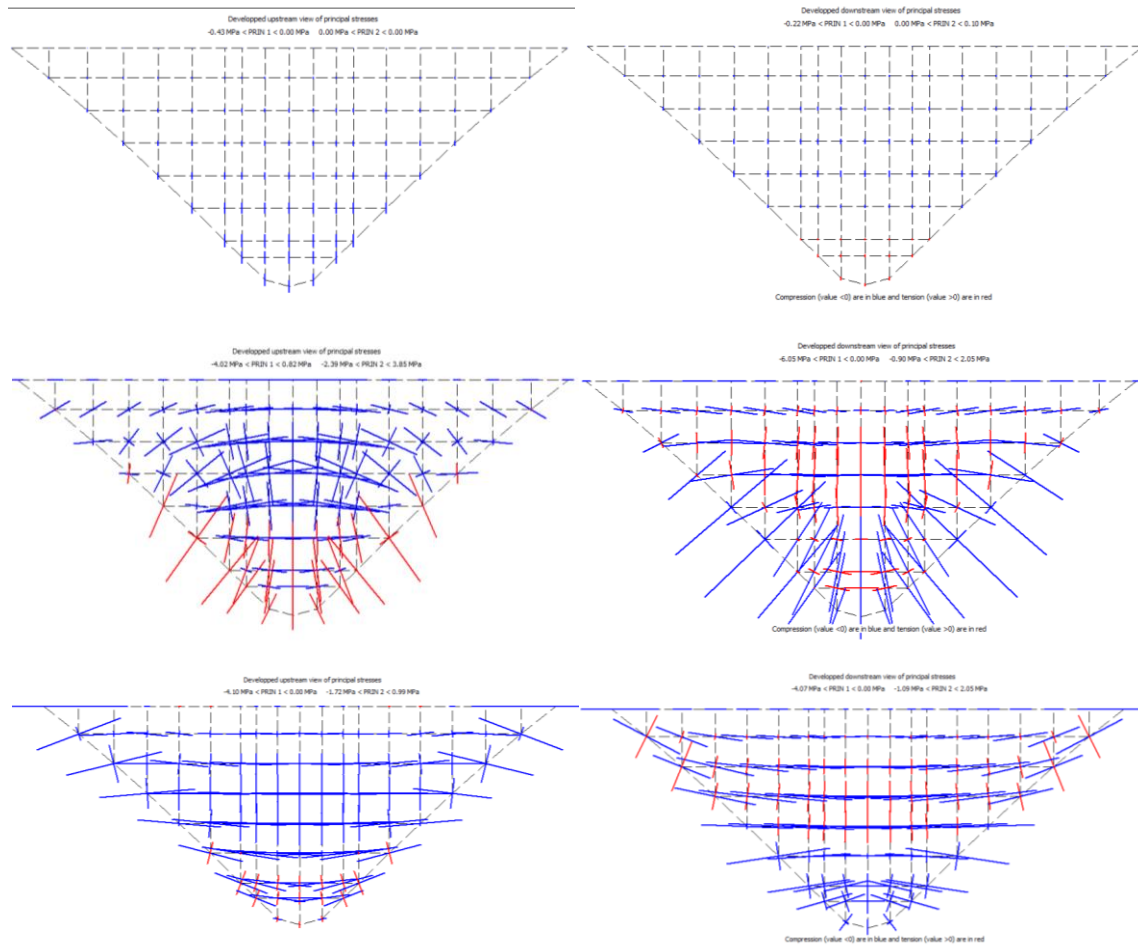


Figure 4. Principal stresses on the upstream (left) and downstream (right) faces for self-weight only load (top), self-weight + hydrostatic loads (middle), self-weight + hydrostatic +10°C thermal increase loads (bottom). Compression in blue, tension in red

VERIFICATION OF ADSAS RESULTS

Assessment of the ADSAS results have been conducted by Salamon and Robbe using a software-to-software verification technique for the Stevenson Creek Dam model (Salamon et.al 2023).

In 1926, a 60-foot-high concrete arch dam was built for the exclusive purpose of validating various analysis methods used in design of arch dams at that time (Engineering Foundation 1927). The dam became known as the Stevenson Creek Experimental Dam and is shown in Figure 5.

As a part of investigations, a structural analysis of the Stevenson Creek Dam was conducted by the authors of the paper using three programs ADSAS, Code_Aster and Diana FEA for a “base conceptual model” that corresponded to one of test case with 40-foot-water head defined.

Four computational models were developed in a linear elastic analysis (Figure 6) as follows:

- Model 1 with elastic (M1) and rigid (M1R) foundation using ADSAS. The dam geometry was represented by 8 arches and 17 cantilevers.
- Model 2 (M2) with elastic foundation using Code_Aster. The dam and foundation were modeled with over 8200 quadratics 20-node hexahedron elements and with total 24 500 nodes.
- Model 3 (M3) with elastic foundation using Diana FEA software (Diana FEA 2022) using over 66 000 and almost 140 000 solid 8-node elements for modeling the dam structure and the foundation, respectively.
- Model 4 with rigid (M4R) and hinged (M4H) boundary conditions at the parameter of a shell dam model using Diana FEA program with 1240 shell elements of 0.3 m maximum size.



Figure 5. Stevenson Creek Experimental Dam during testing, California 1926 (Southern California Edison Photographs and Negatives, Huntington Digital Library)

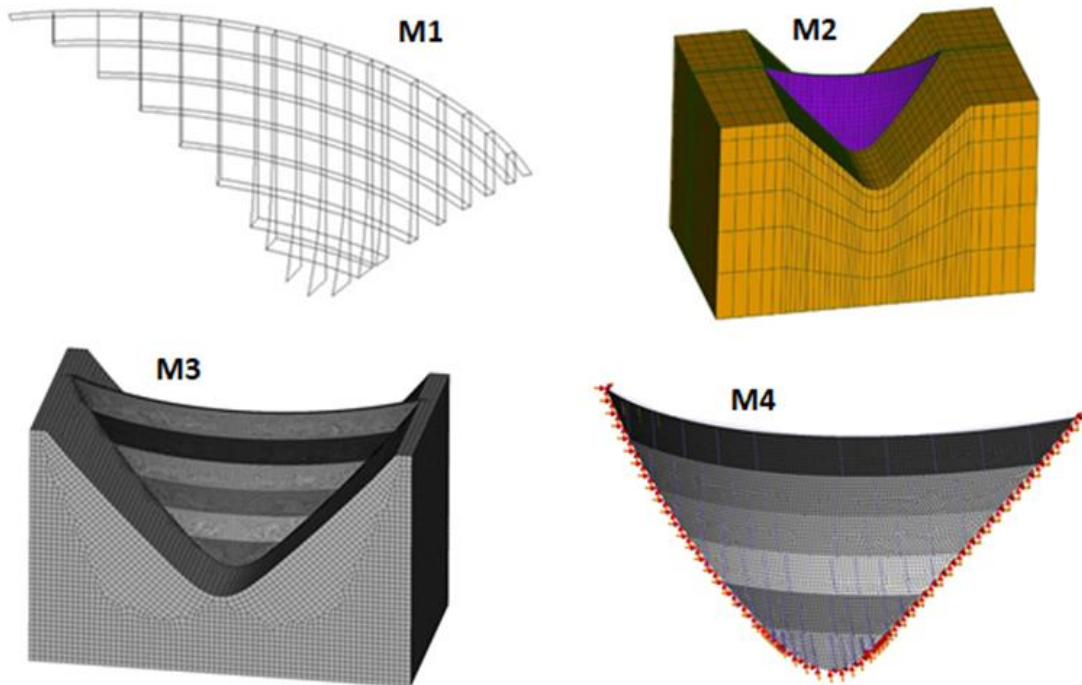


Figure 6. Computational models: M1(ADSAS); M2 (Code_Aster); M3 (Diana Solid); M4 (Diana Shell).

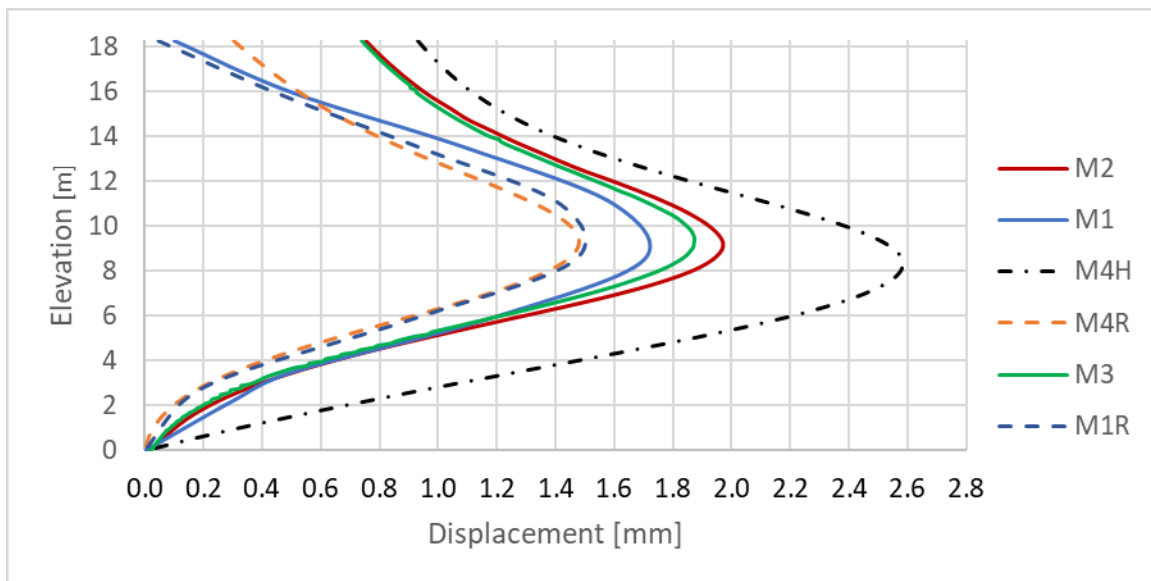


Figure 7. Displacement at the central cantilever for the “base conceptual model”

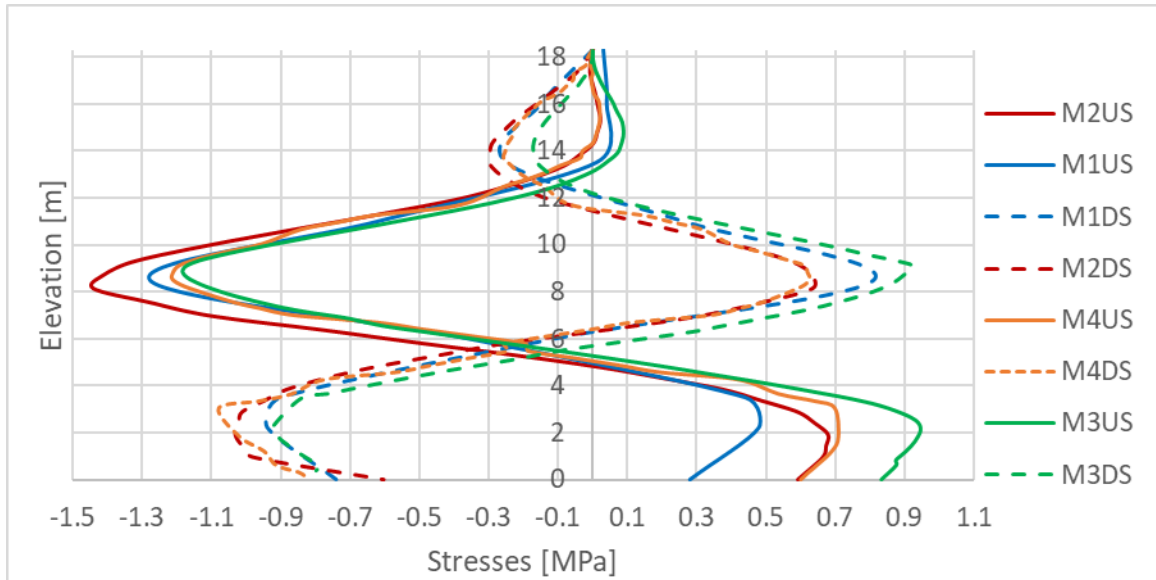


Figure 8. Vertical stresses at upstream and downstream faces of the central cantilever

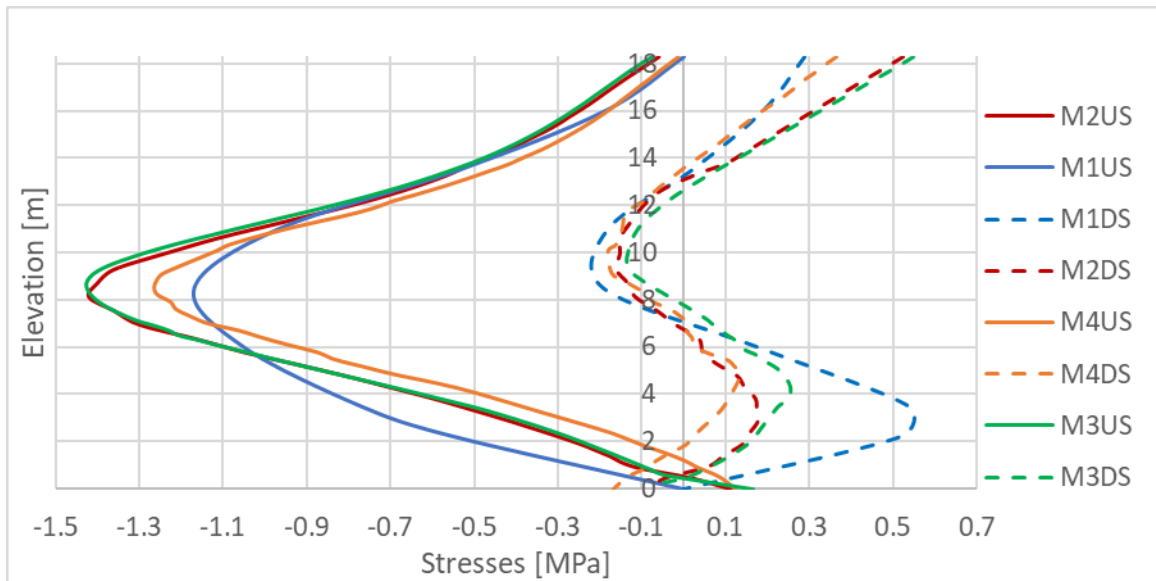


Figure 9. Horizontal stresses at the upstream and downstream face of the central cantilever

Figure 7 through Figure 9 present the comparison of results considering self-weight and 40 feet of water head with the different models. Overall, a good agreement in displacements and in vertical and horizontal stresses at the central cantilever was achieved for ADSAS and FE models. The maximum displacement obtained from ADSAS (M1) are in a good agreement with solid element models M2 and M3 results at the middle height for an elastic foundation model but are significantly smaller at the dam crest.

For a rigid foundation, a similar displacement distribution is observed between the ADSAS model M1R and shell element model M4R. However, the overall magnitude of displacements for rigid foundations is smaller when compared with models with elastic

foundation model. When hinged boundary conditions are applied to the shell element in model M4H, the maximum displacement is significantly increased.

Relatively similar distributions of vertical and horizontal stresses are observed at the dam central cantilever at the upstream and downstream face for ADSAS and the FE analyses.

Figure 10 compares the principal stress distributions computed with ADSAS and Code_Aster finite-element analyses for the full water load (previous results were computed with only 2/3 of the water load). Relatively similar distributions of the principal stresses are computed on upstream and downstream face of the dam.

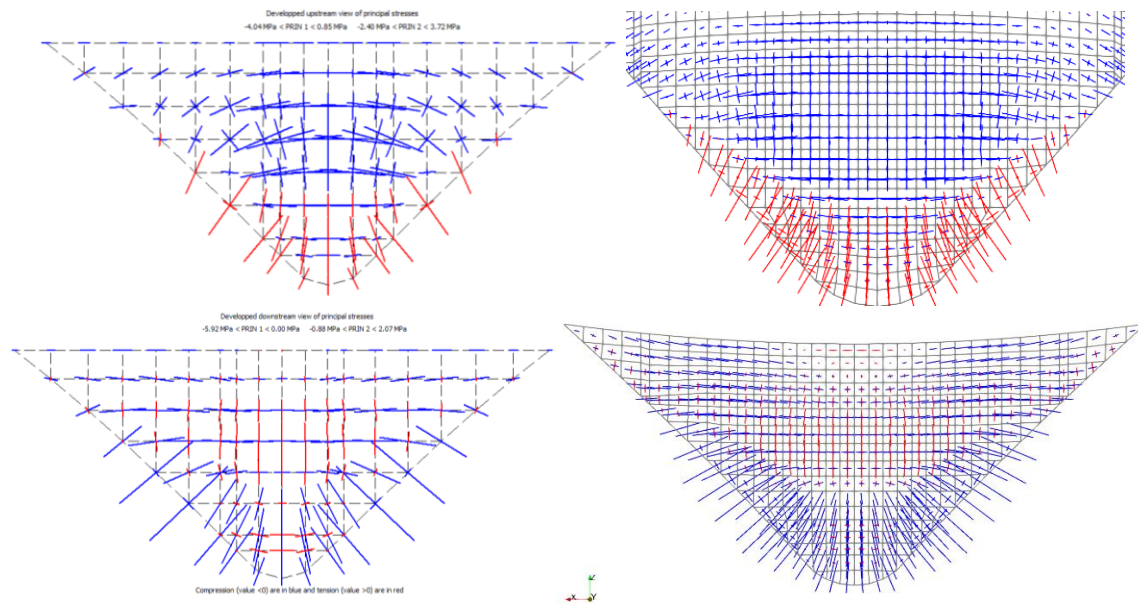


Figure 10. Principal stresses computed by ADSAS (left) and FE (right) on the upstream (top) and downstream (bottom) faces for full water load.

CONCLUSION

In current practice, analyses of arch dams are conducted generally using 3D FE analysis that require significant effort to build and run complex models. The authors demonstrated in the paper, that the Trial-Load Method approach implemented in the ADSAS software is a valuable and efficient tool for verification of the advanced computations. The intuitive nature of the Trial-Load Method helps to better understand and imagine distributions of forces in arch dams and an overall behavior of the structures.

From the authors point of view, the Trial-Load Method and the concept of loads allocated to arches and cantilevers remains a great tool to understand the behavior of arch dams. Even if fewer arch dams are built nowadays, the existing one still require experienced engineers to monitor and understand their behavior. While many of today's dam experts started with Trial-Load Method, such tools are now rarely available for the education of young engineers.

In this spirit, the authors are offering an updated version of the ADSAS software for educational purpose. The program with a user-friendly graphical interface can be run on any personal computers with Windows OS. This updated version of ADSAS is also an opportunity to keep alive this great achievement in the dam engineering history.

ACKNOWLEDGEMENT: The authors would like to thank Mrs. Hillery Venturini at the US Bureau of Reclamation for valuable input and suggestions made during preparation of the paper.

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